

MEHDI BELHAJ

Power Electronics & Hardware Engineer

Analog & digital electronics · Embedded systems · Power conversion · Modelling & control · Aerospace qualification — Toulouse & Europe

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270 VDC

SiC 1200 / 1700 V

10 kW

DO-160G · DAL A

EMC / EMI

OrCAD · KiCad

Matlab · Simulink · Simscape

PI / PID controllers

PSIM · LTspice

C · C++ · Python

AVAILABILITY

Immediate

EXPERIENCE

3 years · aerospace

MOBILITY

Toulouse · France · Europe

WORK PERMIT

Valid to Nov. 2028

DRIVING LICENCE

B

ENGLISH

C1 · fluent

CAREER OBJECTIVE

Join an engineering team on a **long-term permanent contract**, ideally as **experienced engineer** in power electronics, analog electronics or control, to design high-criticality embedded boards and systems covering the full V-cycle, from specification to qualification testing. Primary target: aerospace and space in the Toulouse area; open to energy, rail, electric mobility and industry, in France and abroad. Longer term, move towards **embedded systems** and software (C / C++ / Python), including **embedded AI**.

PROFILE SUMMARY

Power electronics and hardware engineer with **3 years of experience** in the aerospace sector (Airbus, Safran Ventilation Systems) and in high-voltage R&D (SuperGrid Institute). Specialised in the complete design of high-voltage embedded boards: **Flyback 540 V, Buck, SEPIC and LDO** supplies, three-phase **SiC 1200 V inverter up to 10 kW**, **DO-160G** EMC filtering (Section 21, categories B and H), analog and digital sensing chains, up to **DAL A** criticality.

Dual hardware and control competence: electrical and electromechanical equations, block-diagram modelling in **Matlab / Simulink / Simscape**, control-law design and **PI / PID controller tuning**, closed-loop simulation and stability analyses (Bode, Nyquist, Nichols, margins) with Control System Toolbox and Simulink Control Design.

Key differentiator: autonomous coverage of the V-cycle as **author of the key deliverables** — design justification file, PCB layout specification, technical requirements specification, hardware verification procedure, test plans, compliance matrix — and active across the industrial process: Windchill PLM, codification and bills of materials, supplier consultations, obsolescence, multidisciplinary meetings and gate reviews.

Profile broadened by **330 h of Cisco / OpenEDG certificate training** in C++ and Python (2025-2026) and by an **entrepreneurial activity** run in parallel (CE, RoHS, WEEE, EPR and UKCA product compliance, sourcing, business management). Bilingual French, native Arabic, English C1 (weekly technical meetings with Safran USA).

KEY STRENGTHS

- **Autonomy across the V-cycle** — sole electronics engineer on my last two assignments, from kick-off to testing, author of all technical deliverables.
- **Dual power & analog culture** — supply chains (Flyback, Buck, SEPIC, LDO), SiC 1200 / 1700 V stages up to 10 kW, sensor conditioning and 24-bit acquisition.
- **Modelling & control** — equations, Simulink / Simscape block diagrams, PI / PID design and tuning, closed-loop simulation, small-signal stability; 130 ms dynamic requirement validated at Airbus.
- **EMC and aerospace qualification** — common- and differential-mode filtering, lightning, ESD, DO-160G campaigns run end to end, dielectric tests up to 20 kV.
- **Project management and supply chain** — multidisciplinary meetings, gate reviews, codification and BOM, supplier consultations, obsolescence and second sourcing, Windchill PLM.
- **Software & data openness** — certified C++ and Python (330 h), oriented towards real-time embedded software, test automation and measurement post-processing.
- **Immediate availability & mobility** — available without notice, mobile in Toulouse, France and Europe.

DETAILED TECHNICAL SKILLS

POWER ELECTRONICS

Three-phase **SiC 1200 V / 1700 V — 10 kW** pulsed inverter · Quasi-resonant Flyback 540 V to 24 V without optocoupler · Isolated multi-output Flyback 28 V to 3.3 V · Buck · SEPIC · LDO · Isolated gate drivers UCC21750 (DESAT, Miller clamp) · Driver supply and biasing · RCD snubbers · DC-bus pre-charge and inrush limiting · Magnetic component sizing

ANALOG ELECTRONICS & SENSING

Current sensing (ACS37002, INA190, shunts) · High-voltage sensing (resistive dividers and MAX1363 ADC) · Temperature (TMP1075) · 24-bit force measurement chain (ADS1247) · Position interface (CAM312) · Operational amplifiers (OPA387) · Signal conditioning and filtering · Galvanic isolation (ISO1640)

DIGITAL & EMBEDDED

TMS320F280049 DSP (C2000) · PIC16F877x · Embedded C · **C++** (OOP, templates, STL) · **Python** — 330 h certified · VHDL (basics) · LabVIEW · UART, SPI, I2C, RS422, PWM, ADC, EEPROM · Contribution to **DAL A** embedded software (equipment software, sequencer, BITE, sub-assembly specifications) · Git · Linux · PyTest

MODELLING, CONTROL & SIMULATION

Matlab / Simulink / Simscape · Control System Toolbox · Simulink Control Design (automated linearisation: sLinearizer, getIOTransfer, getLoopTransfer, getSensitivity) · Scilab / Xcos · Electrical and electromechanical equations (converters, DC bus, motor, actuator) · Transfer functions and **block-diagram modelling** · **Control-law** design · **PI / PID controller** design and tuning (identification, pole placement, margin criteria) · Nested current / voltage loops · **Closed-loop simulation**: load steps, field weakening · Small-signal stability (open-loop transfer functions, sensitivities, gain, phase, delay and modulus margins) · Bode, Nyquist, Nichols · PSIM (FFT) · LTspice · Saber

EMC, PROTECTIONS & TESTING

DO-160G Section 21 — categories B and H: conducted and radiated emissions, LISN measurements · Common- and differential-mode filters · Lightning protection by pin injection · 15 kV ESD · Reverse-polarity protection · Phase-loss detection on ± 270 VDC · Over-voltage and over-temperature · Dielectric tests up to 20 kV · Insulation measurements · IEC 60068 · ABD0100 requirements · Bench debugging, oscilloscope

CAD & PCB DESIGN

OrCAD Capture CIS · KiCad · Proteus · **IPC-2221** layout specifications (high-voltage clearances corrected for altitude per IEC 60664-1) · 6-layer FR4 PCB, stack-up · Current-based trace sizing · DFX and testability · EMC and thermal constraints · Mechanical integration on 3D model · Flex circuits · IPC-A-600, IPC-A-610, IPC-7351

PROCESS, QUALITY & INDUSTRIALISATION

Full V-cycle · Planning, milestone and deadline tracking · Interface with Safety and quality teams · Internal and gate reviews · Authoring: technical requirements specification, hardware verification procedure, design justification file, calculation notes, test plans, compliance matrix, lessons-learned sheet · **Windchill PLM**, codification and bills of materials · Supplier consultations · Aerospace component qualification · Obsolescence and second sourcing · **DO-254** (basics) · CE, RoHS, WEEE, EPR, UKCA product compliance · International customer interface in English

DEVELOPMENT METHODOLOGY

Specification

Requirements · TRS · DO-160 · ABD0100

Architecture

Block diagram · topologies · budgets

Design

Schematics · DJF · modelling · layout · BOM

Build

Prototype boards · procurement

Unit tests

Voltages · insulation · bench debug

Integration

EMC · lightning · ESD · thermal

Validation

Compliance matrix · capitalisation

DETAILED PROFESSIONAL EXPERIENCE



Power Electronics & Hardware Engineer — Safran Ventilation Systems

Via Centum T&S · Toulouse (Blagnac) · Embedded 270 VDC power conversion · DO-160 category H

Nov. 2023 — May 2025 · 19 months

CONTEXT

Adaptation of an existing aircraft ventilation system from the 115 VAC network to the **270 VDC** high-voltage network (more-electric aircraft and helicopter), to meet the needs of future helicopter programmes. Goal: deliver a commercially oriented demonstrator while minimising software and mechanical impacts. Sole electronics engineer on the project, with the V-cycle covered autonomously.

ACHIEVEMENTS

- **Architecture & hardware design**: complete design of the **input board** (EMI filter and power supplies) within a multi-board architecture (input, inverter control, interface, power); 1,890 W delivered, -40 to $+100$ °C temperature range.
- **Nine hardware functions** designed and justified: diode reverse-polarity protection; low-voltage supplies (isolated Flyback, 18 / 16 / 5 V LDOs, Buck converter) with separate rails for the high- and low-voltage groups; **phase-loss detection in under 4 ms** with supply hold-up over micro-cuts of 200 ms and more; inrush limiting to a normative envelope ($9\times / 4\times / 2\times$ nominal current); lightning protection by pin injection; 15 kV ESD; category H EMI filter; over-temperature protection; temperature measurement.
- **PCB design & layout**: layout specification — **6-layer FR4 PCB** (35 μ m, 1.6 mm, class 4, selective NiAu finish), stack-up definition, **IPC-2221** clearance calculations corrected for altitude per IEC 60664-1, DC-bus traces rated for 9 A RMS ($\Delta T < 10$ °C), Y-shaped lightning routing on outer layers, DFX rules and testability (100 % of nets testable).
- **Mechanical integration**: interface drawing, heat-exchange zones, connector placement, iterations with the mechanical design office on the 3D model.
- **System simulation & EMC qualification**: full inverter, motor and control system modelled in PSIM, control tests in simulation, load-step responses, FFT analyses; DO-160G category H — common- and differential-mode filter sizing, RCD snubbers; EMI test plans written and executed (LISN, conducted and radiated emissions); lightning tests and insulation measurements.
- **Industrial documentation**: **design justification file** with per-function justification and calculation notes, **technical requirements specification**, **hardware verification procedure**, **requirements verification matrix** and lessons-learned sheet.
- **Configuration management, PLM & procurement**: component codification and bills of materials in Windchill; supplier consultations and selection, aerospace component qualification (temperature, altitude, derating), obsolescence and second sourcing.
- **Organisation & coordination**: activity planning and milestone management (design reviews, test campaigns), deadline and deliverable tracking, multidisciplinary technical meetings (hardware, layout, mechanical, procurement), internal and gate reviews, weekly technical meeting led in English with the Safran USA teams.

Results: functional mock-up validated in EMI tests on the applicable DO-160 sections; complete design package delivered; 270 VDC feasibility demonstrated with minimised software and mechanical impacts, work capitalised for future 270 VDC programmes.

Environment: OrCAD Capture CIS · KiCad · PSIM · LTspice · Matlab/Simulink · TMS320F280049 · DO-160G cat. H · IPC-2221 / A-600 / A-610 / 7351 · IEC 60664-1 · IEC 60068 · Windchill PLM

CONTEXT

Proof of concept of an ultra-fast electromechanical actuator ("Pop-Up Spoiler" EMA) deploying a wing surface in **130 ms** or less, to counter gusts and lighten the wing structure, within the Zero Emission aircraft studies. Design of the motor-control electronics: three boards — 28 V CPU, ± 270 V HVDC and inverter — in a constrained $65 \times 65 \times 170$ mm volume. Partnership between Serma Ingénierie, Crouzet and Airbus.

ACHIEVEMENTS

- **Requirements, specifications & safety:** contribution to the hardware and software requirements; interface with the Safety team (Serma Safety) for the safety assessment and the flow-down of ABD0100 aerospace requirements.
- **SiC driver board:** supply, biasing and gate driving of 6 SiC 1200 V MOSFETs — isolated UCC21750 drivers with DESAT protection, four driver supplies, phase-current and temperature sensors; three-phase inverter delivering **10 kW in pulsed operation**.
- **HVDC board:** ESD protection and inrush limiting, 540 V EMC filter (common and differential modes, 900 V ceramic capacitors), high-voltage current and voltage sensing (ACS37002 and MAX1363 ADC), **quasi-resonant Flyback 540 V to 24 V without optocoupler** (auxiliary-winding regulation) and Buck 24 V to 12 V for the brake and locking device (TPS51604, INA190 sensing).
- **CPU board:** 28 V EMC filter, ESD and reverse-polarity protection, dual-output Flyback 28 V to 3.3 V analog / digital (LM5155), 5 V converter, force acquisition chain with a **24-bit ADS1247 ADC**, CAM312 position interface, **TMS320F280049 DSP** and peripherals (UART, SPI, I2C, RS422, PWM, ADC).
- **Modelling & control laws:** electrical and electromechanical equations of the sub-assemblies (motor, power electronics, mechanical chain), **block-diagram models in Matlab / Simulink / Simscape** and Scilab / Xcos, parameter identification; control-law design and **controller tuning**, closed-loop simulation and validation of the 130 ms dynamic requirement (Bode and Nyquist analyses); full inverter, motor and control system modelled in PSIM — control tests, load-step responses, FFT analyses — and LTspice simulations (PWM, losses, transients).
- **DAL A embedded software:** contribution to the preliminary and detailed design of the equipment software, sequencer and built-in tests (BITE); specifications of the electronic sub-assemblies.
- **Reliability:** junction-temperature analysis of the SiC components in pulsed operation (thermal shocks linked to the 130 ms sequences), ageing risks, reliability reviews with the partners.
- **Organisation, integration & reviews:** planning of the electronics activities, milestone and deadline tracking, coordination of the hardware, software, mechanical and safety teams; board-to-board flex links, dedicated driver-supply board, work on the 3D model with the mechanical engineers; preliminary and detailed design justification files; design reviews with Airbus and Crouzet.

Results: 130 ms requirement validated in simulation; hardware architecture approved in Airbus and Crouzet reviews; proof of concept integrated into the Zero Emission studies; know-how capitalised on transient control.

Environment: OrCAD · KiCad · LTspice · PSIM · Matlab / Simulink / Simscape · Scilab / Xcos · SiC 1200 V · UCC21750 · BM2SC123FP2 · LM5155 · TPS51604 · MAX1363 · INA190 · ADS1247 · CAM312 · ACS37002 · TMP1075 · DAL A · ABD0100 · DO-160G Section 21



Graduate internship — R&D Power Electronics Engineer — SuperGrid Institute

Feb. — Jul. 2022 · 6 months

Villurbanne (Lyon) · Research institute · High-voltage DC grids (HVDC)

CONTEXT

Research and development on high-voltage DC grids: design of an isolated high-voltage supply for the gate drivers of 6 series-connected 1.7 kV SiC MOSFETs in a high-voltage leg.

ACHIEVEMENTS

- **Design:** isolated HV supply based on 12 cascaded ferrite transformers driving 6 series-connected 1.7 kV SiC MOSFETs (high and low side); TDK EPCOS T38 core selection balancing losses, insulation and footprint.
- **Simulation & correlation:** LTspice modelling of the transformers' magnetic and electrical behaviour, validated by oscilloscope measurements.
- **EMC analysis:** characterisation of common-mode currents generated by edges up to 16 kV/ μ s and definition of countermeasures.
- **Laboratory testing:** prototyping and dielectric tests up to 20 kV to validate insulation and safety.
- **Industrialisation:** specification for industrial prototyping (efficiency, EMC compatibility).

Results: 20 kV insulation validated, 12-transformer cascade operational, industrialisation specifications delivered to the prototyping team.

Environment: LTspice · Oscilloscope · 20 kV dielectric tests · TDK EPCOS T38 ferrites · 1700 V SiC MOSFETs · EMC

CONTINUING EDUCATION — SOFTWARE DEVELOPMENT



Continuing education — C++ & Python software development — Cisco Networking Academy / OpenEDG

2025 — 2026 · 330 h

In partnership with OpenEDG C++ Institute and Python Institute · curricula aligned with the CPE / CPA / CPP and PCEP / PCAP frameworks · 5 verifiable Credly badges

CONTEXT

Structured software upskilling (330 h) completed in 2025-2026 alongside the entrepreneurial activity, oriented towards real-time embedded software and test tooling.

ACHIEVEMENTS

- **C++ Advanced (CPP):** function and class templates, full STL (containers, algorithms, sorting, merging, sets, iterators), stream, file and string I/O.
- **C++ Essentials 1 & 2 (CPE, CPA):** types, operators, flow control, functions; pointers and dynamic allocation; complete object-oriented programming (inheritance, polymorphism, abstraction); exceptions, namespaces, streams.

- **Python Essentials 1 & 2 (PCEP, PCAP):** algorithms, script design and debugging; object-oriented programming, exceptions, iterators and generators, modules, packages and PIP, file handling.
- **Target applications:** real-time embedded software in C / C++, test automation and measurement post-processing in Python (test benches, EMI campaigns, oscilloscope acquisitions).
- **Published projects** — github.com/mehdi-belhaj-eea: scope-emi-toolkit (oscilloscope CSV parser, dBμV spectrum, EMC pre-check, 8 unit tests, continuous integration) · embedded-dsp-cpp (ring buffer and digital filters in C++17, 12 GoogleTest tests).

Results: 5 verifiable Credly badges, CPP / CPA / CPE and PCAP / PCEP frameworks.



ENTREPRENEURSHIP — IN PARALLEL

TOKYOSEAT

Co-founding partner — TOKYOSEAT
Electrical and sanitary equipment · SAS with 5 partners · trademark registered at INPI · Amazon FBA and Shopify (France, Germany, Italy, Spain, Netherlands) and United Kingdom

Since Jul. 2023 · ongoing

CONTEXT

Creation and development of a European brand of electrical and sanitary home equipment, distributed mainly on European marketplaces. Founding-partner role covering product compliance, sourcing, supply chain and business management. The business is now stable, with day-to-day operations run by an employee: compatible with a full-time engineering commitment.

ACHIEVEMENTS

- **Product & regulatory compliance:** CE marking under the Low Voltage and EMC directives, RoHS, WEEE, extended producer responsibility (EPR) and UKCA — laboratory test coordination (electrical safety, EMC), preparation and verification of technical files and EU declarations of conformity with French and German bodies, multi-country regulatory watch.
- **Environmental compliance & labels:** WEEE and packaging EPR registrations in France and Germany (Stiftung EAR register); German quality label “SEHR GUT (1.3)” obtained on the flagship product (DasQualitätssiegel.de, independent product tests, October 2025).
- **Supplier & product quality:** factory selection, negotiation, sampling campaigns, quality control, improvement plans built from customer feedback; filing and monitoring of the TOKYOSEAT trademark at INPI (classes 9 and 11).
- **Operations & automation:** Amazon FBA (pan-European programme) and Shopify store backed by a third-party fulfilment centre (storage, order preparation, delivery); import (EORI, TARIC classification, customs, VAT, forwarders); automation of order flows and customer service, API integrations, automated report and KPI analysis.
- **Business management:** business plans, cost price, margins, ROI, pricing scenarios, cash flow, profitability dashboards per SKU; recruitment and management of one employee.

Results: compliance maintained on 6 markets (EU and UK) without interruption of sales; German quality label obtained; German and British markets developed in 2025-2026; business stable and run day to day by an employee.

KEY MEASURABLE ACHIEVEMENTS

PROJECT	MISSION	RESULT / IMPACT
270 VDC conversion Safran SVS · 2023-25	Input board, 9 protection functions, layout specification, DO-160G cat. H qualification, full documentation, procurement and codification.	EMI-validated mock-up · complete package delivered · V-cycle covered autonomously
EMA actuator Airbus · 2022-23	3 boards within 65 × 65 × 170 mm: 10 kW SiC inverter, 540 V Flyback, Simulink / Simscape modelling and control laws, DAL A software, safety.	130 ms requirement validated · architecture approved in Airbus and Crouzet reviews
Isolated HV SiC supply SuperGrid · 2022	12 cascaded ferrite transformers for 6 series-connected 1.7 kV SiC MOSFETs, EMC analysis at 16 kV/μs.	20 kV insulation validated · industrialisation specifications delivered
TOKYOSEAT since 2023	CE, RoHS, WEEE, EPR and UKCA compliance; sourcing and supplier quality; import and supply chain; business management.	6 compliant markets · “SEHR GUT (1.3)” label
Software training Cisco / OpenEDG · 2025-26	330 h: C++ Advanced, C++ Essentials 1 and 2, Python Essentials 1 and 2.	5 verifiable Credly badges

EDUCATION & CERTIFICATIONS

 	Master's degree in EEA (5-year degree) • Université Toulouse III Paul Sabatier • INP Toulouse – ENSEEIHT Electronics, Electrical Energy, Automatic Control • Electrical Energy track: Conversion, Materials, Sustainable Development • specialisation in Power Electronics, Actuators and Control	2020 – 2022
	Bachelor's degree in EEA — Electronics, Electrical Energy, Automatic Control • Université Toulouse III Paul Sabatier	2019
	Two-year technical degree (DUT) in Electrical Engineering and Industrial Computing • ENSAM Rabat, Morocco	2018
	Cisco Networking Academy / OpenEDG — C++ and Python C++ Advanced, C++ Essentials 1 and 2, Python Essentials 1 and 2 • verifiable Credly badges	2025 – 2026 • 330 h
	French electrical safety authorisation H0, B2V, B2V Essai, BE Essai — CAMIRA Formation Prévention Non-electrical work near HV installations (H0), work and testing on LV installations (B2V, BE Essai) • refresher due	2022
  	EYE Berlin — European Youth Event • European Parliament & Schwarzkopf Foundation • certificate of participation Berlin, 18-20 April 2024: 1,500 young Europeans gathered for workshops and debates on the European elections and the future of a sustainable, innovative and united Europe	Apr. 2024

LANGUAGES & SOFT SKILLS

French — bilingual • **Arabic** — native • **English** — C1, fluent: weekly technical meetings with Safran USA, documentation and deliverables in English • **Spanish, Portuguese** — basics

Autonomous • rigorous • curious • responsive • team player • proactive • Competitive football • NQT mentoring • travel — 18 countries visited

CONDITIONS & AVAILABILITY

Availability: immediate, no notice period • **Contract:** permanent, fixed-term, project-based or temporary — full time

Mobility: Toulouse, Occitanie, all of France, Europe — occasional travel accepted • **Driving licence:** B

Status: Moroccan national, French salaried residence permit valid until November 2028

Target sectors: aerospace and space, energy, rail, electric mobility, semiconductors, industry

Skills portfolio — Mehdi Belhaj • document prepared for recruitment purposes: all information corresponds to assignments actually carried out • References available on request • Updated — October 2026